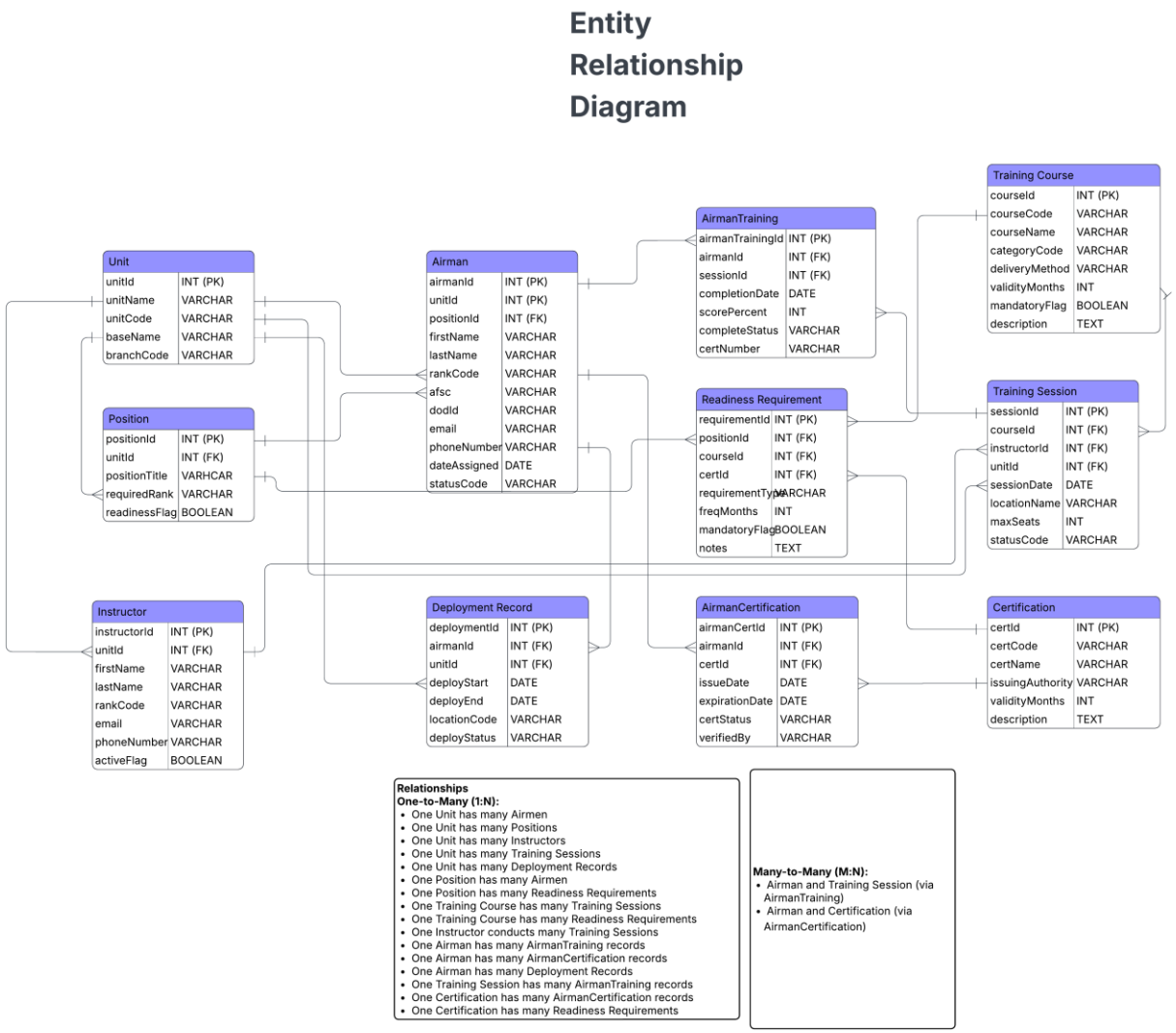
Entity-Relationship Diagram (ERD)



Data Dictionary

Unit Table

P/F	Field Name	Description	Data Type	Size	Notes
P	unitId	Unique unit identifier	INT	-	PRIMARY KEY
	unitName	Name of the unit	VARCHAR	100	NOT NULL
	unitCode	Code used to identify the unit	VARCHAR	20	UNIQUE

	baseName	Base where unit is located	VARCHAR	100	
	branchCode	Military Branch identifier	VARCHAR	10	

Position Table

P/F	Field Name	Description	Data Type	Size	Notes
P	positionId	Unique identifier for each position	INT	-	PRIMARY KEY
F	unitId	Unit that the position belongs to	INT	-	FOREIGN KEY Unit (unitId)
	positionTitle	Title of the assigned position/billet	VARCHAR	100	NOT NULL
	requiredRank	Rank normally required for position	VARCHAR	10	
	readinessFlag	Indicates whether position is readiness critical	BOOLEAN	-	DEFAULT FALSE

Airman Table

P/F	Field Name	Description	Data Type	Size	Notes
P	airmanId	Unique identifier for each airman	INT	-	PRIMARY KEY
F	unitId	Unit the airman is assigned to	INT	-	FOREIGN KEY Unit(unitId)
F	positionId	Position assigned to airman	INT	-	FOREIGN KEY Position(positionId)
	firstName	Airman’s first name	VARCHAR	50	NOT NULL
	lastName	Airman’s last name	VARCHAR	50	NOT NULL
	rankCode	Airman’s rank	VARCHAR	10	NOT NULL
	afsc	Air Force Specialty Code	VARCHAR	10	

	dodId	Unique DoD identification number	VARCHAR	20	UNIQUE, NOT NULL
	email	Airman’s email address	VARCHAR	100	
	phoneNumber	Airman’s contact phone number	VARCHAR	20	
	dateAssigned	Date airman was assigned to unit	DATE	-	NOT NULL
	statusCode	Current personnel status	VARCHAR	20	

Instructor Table

P/F	Field Name	Description	Data Type	Size	Notes
P	instructorId	Unique identifier for each instructor	INT	-	PRIMARY KEY
F	unitId	Unit the instructor belongs to	INT	-	FOREIGN KEY Unit(unitId)
	firstName	Instructor’s first name	VARCHAR	50	NOT NULL
	lastName	Instructor’s last name	VARCHAR	50	
	rankCode	Instructor’s rank	VARCHAR	10	
	email	Instructor’s email address	VARCHAR	100	
	phoneNumber	Instructor’s contact phone number	VARCHAR	20	
	activeFlag	Indicates whether the instructor is currently active	BOOLEAN	-	DEFAULT TRUE

TrainingCourse Table

P/F	Field Name	Description	Data Type	Size	Notes
P	courseId	Unique identifier for	INT	-	PRIMARY KEY

		each training course			
	courseCode	Code used to identify the course	VARCHAR	20	UNIQUE, NOT NULL
	courseName	Name of the training	VARCHAR	100	NOT NULL
	categoryCode	Training category or readiness area	VARCHAR	30	
	deliveryMethod	Method of course delivery	VARCHAR	30	
	validityMonths	Number of months training remains valid	INT	-	CHECK > 0
	mandatoryFlag	Indicates whether the course is mandatory	BOOLEAN	-	DEFAULT FALSE
	description	Description of the course	TEXT	-	

TrainingSession Table

P/F	Field Name	Description	Data Type	Size	Notes
P	sessionId	Unique identifier for each training session	INT	-	PRIMARY KEY
F	courseId	Course being taught in the session	INT	-	FOREIGN KEY TrainingCourse(courseId)
F	instructorId	Instructor conducting the session	INT	-	FOREIGN KEY Instructor(instructorId)
F	unitId	Unit hosting or assigned to the session	INT	-	FOREIGN KEY Unit(unitId)
	sessionDate	Date of training session	DATE	-	NOT NULL
	locationName	Location where	VARCHAR	100	

		session is held			
	maxSeats	Maximum available seats	INT	-	CHECK > 0
	statusCode	Current status of the session	VARCHAR	20	

AirmanTraining Table

P/F	Field Name	Description	Data Type	Size	Notes
P	airmanTrainingId	Unique identifier for each Airman training record	INT	-	PRIMARY KEY
F	airmanId	Airman associated with training record	INT	-	FOREIGN KEY Airman(airmanId)
F	sessionId	Training session attended by airman	INT	-	FOREIGN KEY TrainingSession(sessionId)
	completionDate	Date airman completed the training	DATE	-	
	scorePercent	Score earned by airman	INT	-	CHECK 0-100
	completionStatus	Completion result for the training	VARCHAR	20	
	certNumber	Certifiicate number issued	VARCHAR	50	

Certification Table

P/F	Field Name	Description	Data Type	Size	Notes

P	certId	Unique identifier for each certification	INT	-	PRIMARY KEY
	certCode	Code used to identify the certification	VARCHAR	20	UNIQUE, NOT NULL
	certName	Name of certification	VARCHAR	100	NOT NULL
	issuingAuthority	Organization or authority issuing certification	VARCHAR	100	
	validityMonths	Number of months the certification remains valid	INT	-	CHECK > 0
	description	Description of the certification	TEXT	-	

AirmanCertification Table

P/F	Field Name	Description	Data Type	Size	Notes
P	airmanCertId	Unique identifier for airman certification record	INT	-	PRIMARY KEY
F	airmanId	Airman who holds the certification	INT	-	FOREIGN KEY Airman(airmanId)
F	certId	Certification held by the airman	INT	-	FOREIGN KEY Certification(certId)
	issueDate	Date the certification was issued	DATE	-	NOT NULL
	expirationDate	Date the certification	DATE	-	
	certStatus	Current status of the certification	VARCHAR	20	
	verifiedBy	Person or office that	VARCHAR	100	

		verified certification			

ReadinessRequirement Table

P/F	Field Name	Description	Data Type	Size	Notes
P	requirementId	Unique identifier for each readiness requirement	INT	-	PRIMARY KEY
F	positionId	Position that requires the training or certification	INT	-	FOREIGN KEY Position(positionId)
F	courseId	Required training course	INT	-	FOREIGN KEY TrainingCourse(courseId)
F	certId	Require certification	INT	-	FOREIGN KEY Certification(certId)
	requirementType	Type of readiness requirement	VARCHAR	30	
	freqMonths	Required renewal frequency in months	INT	-	CHECK > 0
	mandatoryFlag	Indicates whether	BOOLEAN	-	DEFAULT TRUE
	notes	Additional requirement notes	TEXT	-	

DeploymentRecord Table

P/F	Field Name	Description	Data Type	Size	Notes
P	deploymentId	Unique identifier for each deployment record	INT	-	PRIMARY KEY
F	airmanId	Airman assigned to the deployment	INT	-	FOREIGN KEY Airman(airmanId)

F	unitId	Unit associated with the deployment	INT	-	FOREIGN KEY Unit(unitId)
	deployStart	Deployment start date	DATE	-	NOT NULL
	deployEnd	Deployment end date	DATE	-	CHECK deployEnd >= deployStart
	locationCode	Code for deployment location	VARCHAR	30	
	deployStatus	Current deployment status	VARCHAR	20	

Database Normalization

UNF	1NF	2NF	3NF
ReadinessMaster	**ReadinessMaster1**	**Unit**	**Unit**
airmanId	airmanId	unitId	unitId
airmanName	firstName	unitName	unitName
rankCode	lastName	unitCode	unitCode
afsc	rankCode	baseName	baseName
dodId	afsc	branchCode	branchCode
unitName	dodId		
unitCode	unitName	**Position**	**Position**
baseName	unitCode	positionId	positionId
positionTitle	baseName	unitId*	unitId*
requiredRank	positionTitle	positionTitle	positionTitle
courseCode	requiredRank	requiredRank	requiredRank
courseName	courseCode	criticalFlag	criticalFlag
sessionDate	courseName		
instructorName	sessionDate	**Airman**	**Airman**
completionDate	instructorFirstName	airmanId	airmanId
scorePercent	instructorLastName	unitId*	unitId*
certCode	completionDate	positionId*	positionId*
certName	scorePercent	firstName	firstName
issueDate	certCode	lastName	lastName

expirationDate	certName	rankCode	rankCode
requirementTyp e	issueDate	afsc	afsc
freqMonths	expirationDate	dodId	dodId
deployStart	requirementsTyp e	email	email
deployEnd	freqMonths	phoneNumber	phoneNumber
locationCode	deployStart	dateAssigned	dateAssigned
	deployEnd	statusCode	statusCode
	locationCode		
		TrainingCourse	**TrainingCourse**
		courseId	courseId
		courseCode	courseCode
		courseName	courseName
		categoryCode	categoryCode
		deliveryMethod	deliveryMethod
		validityMonths	validityMonths
		mandatoryFlag	mandatoryFlag
		description	description
		Certification	**Certification**
		certId	certId
		certCode	certCode
		certName	certName
		issuingAuthority	issuingAuthority
		validityMonths	validityMonths
		description	description
		TrainingSession	**TrainingSession**
		sessionId	sessionId
		courseId*	courseId*
		instructorId*	instructorId*
		unitId*	unitId*
		sessionDate	sessionDate
		locationName	locationName
		maxSeats	maxSeats
		statusCode	statusCode
		AirmanTraining	**AirmanTraining**
		airmanTrainingId	airmanTrainingId
		airmanId*	airmanId*
		sessionId*	sessionId*

		completionDate	completionDate
		scorePercent	scorePercent
		completionStatus	completionStatus
		certNumber	certNumber
		AirmanCertification	**AirmanCertification**
		airmanCertId	airmanCertId
		airmanId*	airmanId*
		certId*	certId*
		issueDate	issueDate
		expirationDate	expirationDate
		certStatus	certStatus
		verifiedBy	verifiedBy
		ReadinessRequirements	**ReadinessRequirements**
		requirementId	requirementId
		positionId*	positionId*
		courseId*	courseId*
		certId*	certId*
		requirementType	requirementType
		freqMonths	freqMonths
		mandatoryFlag	mandatoryFlag
		notes	notes
		DeploymentRecord	**DeploymentRecord**
		deploymentId	deploymentId
		airmanId*	airmanId*
		unitId*	unitId*
		deployStart	deployStart
		deployEnd	deployEnd
		locationCode	locationCode
		deployStatus	deployStatus

- * Denotes foreign key

Normalization Summary:

The database design began as a single unnormalized table containing Airman, Unit, Position, Training, Certification, Requirement, and Deployment data. In first normal form, repeating groups were removed and all fields were made atomic. In second normal form, the data was separated into entity-specific tables such as Airman, Unit, Position, TrainingCourse, and Certification to eliminate partial dependencies. In third

normal form, transitive dependencies were removed by using foreign keys and junction tables, resulting in a fully normalized database structure. The final design matches the ER diagram, where each table represents a single entity and all non-key attributes depend only on their primary key.

Data Definition Language (DDL)

```
-- UNIT
CREATE TABLE Unit (
      unitId INT PRIMARY KEY,
      unitName VARCHAR(100) NOT NULL,
      unitCode VARCHAR(20) UNIQUE,
      baseName VARCHAR(100),
      branchCode VARCHAR(10)
);

-- POSITION
CREATE TABLE Position (
      positionId INT PRIMARY KEY,
      unitId INT NOT NULL,
      positionTitle VARCHAR(100) NOT NULL,
      requiredRank VARCHAR(10),
      readinessCriticalFlag BOOLEAN DEFAULT FALSE,
      FOREIGN KEY (unitId) REFERENCES Unit(unitId)
);

-- Airman
CREATE TABLE Airman (
      airmanId INT PRIMARY KEY,
      unitId INT NOT NULL,
      positionId INT NOT NULL,
      firstName VARCHAR(50) NOT NULL,
      lastName VARCHAR(50) NOT NULL,
      rankCode VARCHAR(10) NOT NULL,
      afsc VARCHAR(10),
      dodId VARCHAR(20) UNIQUE NOT NULL,
      email VARCHAR(100),
      phoneNumber VARCHAR(20),
      dateAssigned DATE NOT NULL,
      statusCode VARCHAR(20),
      FOREIGN KEY (unitId) REFERENCES Unit(unitId),
      FOREIGN KEY (positionId) REFERENCES Position(positionId)
```

```
);

-- INSTRUCTOR
CREATE TABLE Instructor (
      instructorId INT PRIMARY KEY,
      unitId INT NOT NULL,
      firstName VARCHAR(50) NOT NULL,
      lastName VARCHAR(50) NOT NULL,
      rankCode VARCHAR(10),
      email VARCHAR(100),
      phoneNumber VARCHAR(20),
      activeFlag BOOLEAN DEFAULT TRUE,
      FOREIGN KEY (unitId) REFERENCES Unit(unitId)
);

-- TRAINING COURSE
CREATE TABLE TrainingCourse (
      courseId INT PRIMARY KEY,
      courseCode VARCHAR(20) UNIQUE NOT NULL,
      courseName VARCHAR(100) NOT NULL,
      categoryCode VARCHAR(30),
      deliveryMethod VARCHAR(30),
      validityMonths INT CHECK (validityMonths > 0),
      mandatoryFlag BOOLEAN DEFAULT FALSE,
      description TEXT
);

-- CERTIFICATION
CREATE TABLE Certification (
      certId INT PRIMARY KEY,
      certCode VARCHAR(20) UNIQUE NOT NULL,
      certName VARCHAR(100) NOT NULL,
      issuingAuthority VARCHAR(100),
      validityMonths INT CHECK (validityMonths > 0),
      description TEXT
);

-- TRAINING SESSION
CREATE TABLE TrainingSession (
      sessionId INT PRIMARY KEY,
      courseId INT NOT NULL,
      instructorId INT NOT NULL,
      unitId INT NOT NULL,
      sessionDate DATE NOT NULL,
      locationName VARCHAR(100),
```

```
      maxSeats INT CHECK (maxSeats > 0),
      statusCode VARCHAR(20),
      FOREIGN KEY (courseId) REFERENCES TrainingCourse(courseId),
      FOREIGN KEY (instructorId) REFERENCES Instructor(instructorId),
      FOREIGN KEY (unitId) REFERENCES Unit(unitId)
);

-- AIRMAN TRAINING (JUNCTION TABLE)
      CREATE TABLE AirmanTraining (
      airmanTrainingId INT PRIMARY KEY,
      airmanId INT NOT NULL,
      sessionId INT NOT NULL,
      completionDate DATE,
      scorePercent INT CHECK (scorePercent BETWEEN 0 AND 100),
      completionStatus VARCHAR(20),
      certNumber VARCHAR(50),
      FOREIGN KEY (airmanId) REFERENCES Airman(airmanId),
      FOREIGN KEY (sessionId) REFERENCES TrainingSession(sessionId)
);


-- AIRMAN CERTIFICATION (JUNCTION TABLE)
CREATE TABLE AirmanCertification (
airmanCertId INT PRIMARY KEY,
airmanId INT NOT NULL,
certId INT NOT NULL,
issueDate DATE NOT NULL,
expirationDate DATE,
certStatus VARCHAR(20),
verifiedBy VARCHAR(100),
FOREIGN KEY (airmanId) REFERENCES Airman(airmanId),
FOREIGN KEY (certId) REFERENCES Certification(certId)
);



-- READINESS REQUIREMENT
CREATE TABLE ReadinessRequirement (
      requirementId INT PRIMARY KEY,
      positionId INT NOT NULL,
      courseId INT,
      certId INT,
      requirementType VARCHAR(30),
      freqMonths INT CHECK (freqMonths > 0),
      mandatoryFlag BOOLEAN DEFAULT TRUE,
      notes TEXT,
      FOREIGN KEY (positionId) REFERENCES Position(positionId),
      FOREIGN KEY (courseId) REFERENCES TrainingCourse(courseId),
```

```
      FOREIGN KEY (certId) REFERENCES Certification(certId)
);

-- DEPLOYMENT RECORD
CREATE TABLE DeploymentRecord (
      deploymentId INT PRIMARY KEY,
      airmanId INT NOT NULL,
      unitId INT NOT NULL,
      deployStart DATE NOT NULL,
      deployEnd DATE,
      locationCode VARCHAR(30),
      deployStatus VARCHAR(20),
      FOREIGN KEY (airmanId) REFERENCES Airman(airmanId),
      FOREIGN KEY (unitId) REFERENCES Unit(unitId),
      CHECK (deployEnd IS NULL OR deployEnd >= deployStart)
);
```